

Electronics Fundamentals — Formula Reference

Every formula comes as a card: the formula and a plain-English meaning on top, then **VARIABLES** (what each symbol is and its unit) and **UNITS & CONVERSION** (what to convert before calculating, with a worked example).

0. Units, Prefixes & Conversions — read this first

Golden rule: formulas expect **base units** — V, A, Ω , F, H, Hz, s, W. Convert prefixed values (4.7 k Ω , 20 mA, 100 nF) to base units, calculate, then convert the answer back.

Prefix	Symbol	Multiply by	Example
giga	G	$\times 1\,000\,000\,000\ (10^9)$	2.4 GHz = 2 400 000 000 Hz
mega	M	$\times 1\,000\,000\ (10^6)$	4.7 M Ω = 4 700 000 Ω
kilo	k	$\times 1\,000\ (10^3)$	4.7 k Ω = 4 700 Ω
(none)	—	$\times 1$	base unit: V, A, Ω , F, H, Hz, s, W
milli	m	$\times 0.001\ (10^{-3})$	20 mA = 0.02 A
micro	μ (often typed “u”)	$\times 0.000\,001\ (10^{-6})$	100 μ F = 0.0001 F
nano	n	$\times 0.000\,000\,001\ (10^{-9})$	100 nF = 0.000 000 1 F = 0.1 μ F
pico	p	$\times 10^{-12}$	22 pF = 0.000 000 000 022 F

Between prefixes: each step G $\rightarrow$ M $\rightarrow$ k $\rightarrow$ (base) $\rightarrow$ m $\rightarrow$ μ $\rightarrow$ n $\rightarrow$ p is $\times 1000$. To a smaller prefix multiply (1.5 k Ω = 1500 Ω ; 0.1 μ F = 100 nF); to a bigger one divide (4700 Ω = 4.7 k Ω).

When do you actually need to convert?

Same kind of quantity with the same prefix: calculate directly — the answer keeps that prefix.

NO conversion needed — combining like with like	CONVERT FIRST — mixed prefixes or different quantities
Series resistors: 1 k Ω + 2 k Ω = 3 k Ω Parallel resistors: (1 k Ω $\times$ 2 k Ω) / (1 k Ω + 2 k Ω) = 0.667 k Ω Capacitors: 10 μ F + 22 μ F in parallel = 32 μ F; 10 μ F and 10 μ F in series = 5 μ F Inductors: 10 μ H + 22 μ H in series = 32 μ H Ratios (divider, dB, duty cycle): units cancel completely	Mixed prefixes: 1 k Ω + 470 Ω $\rightarrow$ 1000 Ω + 470 Ω = 1470 Ω (not 471!) 1 μ F + 470 nF $\rightarrow$ 1000 nF + 470 nF = 1470 nF Different quantities multiplied or divided ($V = I \times R$, $\tau = R \times C$, X_C , f_0): prefixes don't cancel — convert to base units or use a shortcut pair below.

Shortcut unit pairs (no conversion needed if you stay inside a pair)

You combine	You get	Example
V $\div$ mA	k Ω	5 V $\div$ 20 mA = 0.25 k Ω = 250 Ω
V $\div$ k Ω	mA	12 V $\div$ 4 k Ω = 3 mA
mA $\times$ k Ω	V	2 mA $\times$ 4.7 k Ω = 9.4 V
V $\times$ mA	mW	5 V $\times$ 200 mA = 1000 mW = 1 W
k Ω $\times$ μ F	ms	10 k Ω $\times$ 100 μ F = 1000 ms = 1 s
M Ω $\times$ μ F	s	1 M Ω $\times$ 10 μ F = 10 s
mH $\div$ k Ω	μ s	10 mH $\div$ 1 k Ω = 10 μ s
1 $\div$ ms / 1 $\div$ μ s	kHz / MHz	1 $\div$ 0.5 ms = 2 kHz

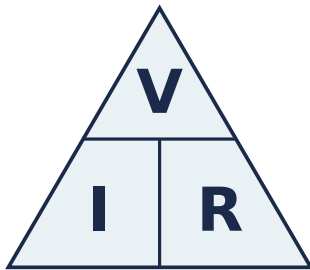
Part markings you will meet

R/k/M notation: the letter is the decimal point — 4k7 = 4.7 k Ω , 2M2 = 2.2 M Ω , R47 = 0.47 Ω . **3-digit codes:** two digits + number of zeros. Resistor “103” = 10 k Ω . Capacitors are in pF: “104” = 100 000 pF = 100 nF; “472” = 4.7 nF.

1. Ohm's Law & Power

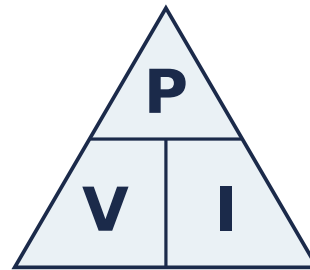
Ohm's law and power triangles — cover the value you want to find

Ohm's law



Cover V → $V = I \times R$
 Cover I → $I = V / R$
 Cover R → $R = V / I$

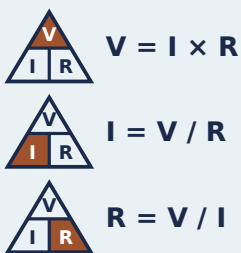
Power



Cover P → $P = V \times I$
 Cover V → $V = P / I$
 Cover I → $I = P / V$

Two letters **side by side** → multiply. One **above** the other → divide top by bottom.

Ohm's law



Relates voltage, current and resistance. Knowing any two gives the third.

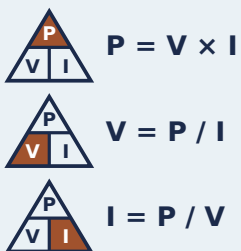
VARIABLES

V — voltage across the component, in volts (V)
I — current through the component, in amps (A)
R — resistance of the component, in ohms (Ω)

UNITS & CONVERSION

Convert mA → A ($\div 1000$) and k Ω → Ω ($\times 1000$).
 V: $0.02 \text{ A} \times 4700 \Omega = 94 \text{ V}$
 I: $9 \text{ V} \div 1500 \Omega = 0.006 \text{ A} = 6 \text{ mA}$
 R: $3 \text{ V} \div 0.015 \text{ A} = 200 \Omega$
 Shortcuts: mA $\times$ k Ω = V; V $\div$ mA = k Ω ; V $\div$ k Ω = mA.

Power



Rate at which a component converts electrical energy, mostly into heat. Use the form that matches the known values; choose a part rated about 2 $\times$ the result.

VARIABLES

P — power, in watts (W)
V — voltage across the component, in volts (V)
I — current through the component, in amps (A)
R — resistance, in ohms (Ω)

UNITS & CONVERSION

Also: $P = I^2 \times R$ and $P = V^2 / R$ (Ohm's law substituted).
 V $\times$ A = W; V $\times$ mA = mW. Convert mA → A *before* squaring.
 P = V $\times$ I: $5 \text{ V} \times 0.2 \text{ A} = 1 \text{ W}$
 P = $I^2 \times R$: $0.02^2 \times 470 \Omega = 0.188 \text{ W}$ → use a 0.5 W resistor
 P = V^2 / R : $12^2 \div 1000 \Omega = 0.144 \text{ W}$

2. Resistors

Series total

$$R_t = R_1 + R_2 + \dots + R_n$$

Resistors in series add. The same current flows through each of them.

VARIABLES

R_t — total (equivalent) resistance
R₁ ... R_n — the individual resistors in the chain

UNITS & CONVERSION

All values in the same unit — the answer comes out in that unit. Don't mix: $1 \text{ k}\Omega + 470 \Omega = 1000 \Omega + 470 \Omega = 1470 \Omega$ (not 471).

Parallel (two resistors) $R_t = (R1 \times R2) / (R1 + R2)$	Total of exactly two parallel resistors. The result is always smaller than the smaller resistor.		
VARIABLES R_t — total resistance of the pair $R1, R2$ — the two parallel resistors	UNITS & CONVERSION Both in the same unit (both Ω or both k Ω). Example: 1 k Ω 1 k Ω = (1 × 1)/(1 + 1) = 0.5 k Ω = 500 Ω . Two equal resistors always give half.		
Parallel (any number) $1/R_t = 1/R1 + 1/R2 + ... + 1/Rn$	Total of any number of parallel resistors. Each added resistor lowers the total.		
VARIABLES R_t — total resistance $R1 ... Rn$ — the individual parallel resistors	UNITS & CONVERSION Same unit for all. Don't forget the last step — flip the sum to get R_t . Example: 1 k Ω , 2 k Ω , 2 k Ω → 1/1 + 1/2 + 1/2 = 2 → R_t = 1/2 = 0.5 k Ω = 500 Ω .		
Voltage divider $V_{out} = V_{in} \times R2 / (R1 + R2)$	Output voltage of two series resistors, taken across R2. Used to scale a voltage down, e.g. for an ADC input.		
VARIABLES V_{out} — voltage at the midpoint, measured across R2 (to ground), in volts V_{in} — voltage across the whole pair, in volts $R1$ — top resistor (between V_{in} and the midpoint) $R2$ — bottom resistor (between the midpoint and ground)	UNITS & CONVERSION R1 and R2 only need to be in the same unit (the units cancel). V_{out} comes out in the unit of V_{in} . Example: 12 V, R1 = 10 k Ω , R2 = 3.3 k Ω → 12 × 3.3 / 13.3 = 2.98 V. Only accurate if the load on V_{out} is at least ~10× bigger than R2.		
Current divider (two resistors) $I1 = I_t \times R2 / (R1 + R2)$	Share of the total current that flows through R1 in a two-resistor parallel pair. Note: the other resistor (R2) is in the numerator.		
VARIABLES $I1$ — current through R1 I_t — total current entering the pair $R1, R2$ — the two parallel resistors	UNITS & CONVERSION Resistors in the same unit; I1 comes out in the unit of I_t (mA in → mA out). Example: 30 mA into 1 k Ω 2 k Ω → I1 (through 1 k Ω) = 30 × 2/3 = 20 mA; the other 10 mA goes through 2 k Ω .		

Resistor colour code

Colour	Digit	Multiplier	Tolerance
Black	0	×1	—
Brown	1	×10	±1%
Red	2	×100	±2%
Orange	3	×1k	—
Yellow	4	×10k	—
Green	5	×100k	±0.5%
Blue	6	×1M	±0.25%
Violet	7	×10M	±0.1%
Grey	8	—	±0.05%
White	9	—	—
Gold	—	×0.1	±5%
Silver	—	×0.01	±10%

4-band: digit, digit, multiplier, tolerance. Brown-black-red-gold = 1, 0, ×100, ±5% → 1000 Ω = 1 k Ω ±5%. 5-band: digit, digit, digit, multiplier, tolerance. Brown-black-black-brown-brown = 100 × 10 = 1 k Ω ±1%.

3. Capacitors

Series total $1/C_t = 1/C_1 + 1/C_2 + \dots + 1/C_n$	Capacitors in series combine by reciprocals. The total is always smaller than the smallest capacitor. The voltage is divided between them.
VARIABLES C_t — total (equivalent) capacitance $C_1 \dots C_n$ — the individual capacitors in series	UNITS & CONVERSION All values in the same unit (all μF or all nF) — the answer is in that unit. Remember to flip the sum at the end. Example: $10\ \mu\text{F} + 10\ \mu\text{F}$ in series $\rightarrow 1/10 + 1/10 = 0.2 \rightarrow C_t = 1/0.2 = 5\ \mu\text{F}$. Mixed units: $1\ \mu\text{F}$ with $470\ \text{nF} \rightarrow$ convert $1\ \mu\text{F} = 1000\ \text{nF}$ first.
Parallel total $C_t = C_1 + C_2 + \dots + C_n$	Capacitors in parallel add. Each one has the same voltage across it.
VARIABLES C_t — total capacitance $C_1 \dots C_n$ — the individual parallel capacitors	UNITS & CONVERSION Same unit for all. Example: $100\ \text{nF} + 0.1\ \mu\text{F} = 100\ \text{nF} + 100\ \text{nF} = 200\ \text{nF}$.
Charge stored $Q = C \times V$	Charge held by a capacitor at a given voltage.
VARIABLES Q — charge, in coulombs (C) C — capacitance, in farads (F) V — voltage across the capacitor, in volts (V)	UNITS & CONVERSION Convert $\mu\text{F} \rightarrow \text{F}$ by multiplying by 10^{-6} ($100\ \mu\text{F} = 0.0001\ \text{F}$). Example: $100\ \mu\text{F}$ at $12\ \text{V} \rightarrow 0.0001 \times 12 = 0.0012\ \text{C}$. Shortcut: $\mu\text{F} \times \text{V}$ gives microcoulombs ($100 \times 12 = 1200\ \mu\text{C}$).
Energy stored $E = \frac{1}{2} \times C \times V^2$	Energy held by a charged capacitor.
VARIABLES E — energy, in joules (J) C — capacitance, in farads (F) V — voltage across the capacitor, in volts (V)	UNITS & CONVERSION C must be in farads. Example: $1000\ \mu\text{F}$ at $25\ \text{V} \rightarrow 0.5 \times 0.001 \times 25^2 = 0.5 \times 0.001 \times 625 = 0.31\ \text{J}$.
Capacitive reactance $X_C = 1 / (2\pi fC)$	Opposition of a capacitor to AC. It decreases as frequency increases; DC is blocked.
VARIABLES X_C — capacitive reactance, in ohms (Ω) π — 3.1416 (so $2\pi \approx 6.283$) f — signal frequency, in hertz (Hz) C — capacitance, in farads (F)	UNITS & CONVERSION Convert f: $\text{kHz} \times 1000$, $\text{MHz} \times 1\ 000\ 000$. Convert C: $\mu\text{F} \times 10^{-6}$, $\text{nF} \times 10^{-9}$, $\text{pF} \times 10^{-12}$. Example: $100\ \text{nF}$ at $1\ \text{kHz} \rightarrow 1 / (6.283 \times 1000 \times 100 \times 10^{-9}) = 1 / 0.000628 \approx 1592\ \Omega$.
RC time constant $\tau = R \times C$	Time a capacitor takes to charge or discharge through a resistor to 63% of the full change.
VARIABLES τ — tau, the time constant, in seconds (s) R — resistance the capacitor charges through, in ohms (Ω) C — capacitance, in farads (F)	UNITS & CONVERSION $\Omega \times \text{F} = \text{seconds}$. Example: $10\ \text{k}\Omega \times 100\ \mu\text{F} = 10\ 000 \times 0.0001 = 1\ \text{s}$. Shortcut: $\text{k}\Omega \times \mu\text{F}$ gives milliseconds ($10 \times 100 = 1000\ \text{ms} = 1\ \text{s}$).

Charging voltage $V(t) = V_s \times (1 - e^{-t/RC})$	Capacitor voltage at time t while charging from 0 V toward the supply voltage.
VARIABLES V(t) — capacitor voltage at time t, in volts V_s — supply voltage it is charging toward, in volts t — time since charging started RC — the time constant τ (see above) e — Euler's number ≈ 2.718 (the e^x key on a calculator)	UNITS & CONVERSION t and RC must be in the same time unit (both s or both ms) — only their ratio matters. Example: $V_s = 5 \text{ V}$, $RC = 1 \text{ s}$, $t = 2 \text{ s} \rightarrow 5 \times (1 - e^{-2}) = 5 \times (1 - 0.135) = 4.32 \text{ V}$.
Discharging voltage $V(t) = V_0 \times e^{-t/RC}$	Capacitor voltage at time t while discharging from its starting voltage toward 0 V.
VARIABLES V(t) — capacitor voltage at time t, in volts V₀ — voltage at the moment discharge starts, in volts t — time since discharge started RC — the time constant τ	UNITS & CONVERSION Same rule: t and RC in the same unit. Example: 12 V , $RC = 10 \text{ ms}$, $t = 10 \text{ ms} \rightarrow 12 \times e^{-1} = 12 \times 0.368 = 4.4 \text{ V}$.
“Fully” settled $t \approx 5\tau$	After 5 time constants the capacitor is more than 99% charged or discharged.
VARIABLES t — settling time τ — time constant $R \times C$	UNITS & CONVERSION Same unit as τ . Example: $\tau = 1 \text{ s} \rightarrow$ about 5 s to fully charge.

4. Inductors

Series total $L_t = L_1 + L_2 + \dots + L_n$	Inductors in series add (when not magnetically coupled).
VARIABLES L_t — total inductance L₁ ... L_n — the individual inductors	UNITS & CONVERSION Same unit for all: $1 \text{ mH} + 470 \text{ }\mu\text{H} = 1000 \text{ }\mu\text{H} + 470 \text{ }\mu\text{H} = 1470 \text{ }\mu\text{H}$.
Parallel total $1/L_t = 1/L_1 + 1/L_2 + \dots + 1/L_n$	Inductors in parallel combine by reciprocals. The total is smaller than the smallest inductor.
VARIABLES L_t — total inductance L₁ ... L_n — the individual inductors	UNITS & CONVERSION Same unit for all; flip the sum at the end. Example: $10 \text{ }\mu\text{H} \parallel 10 \text{ }\mu\text{H} \rightarrow 1/10 + 1/10 = 0.2 \rightarrow 5 \text{ }\mu\text{H}$.
Inductive reactance $X_L = 2\pi fL$	Opposition of an inductor to AC. It increases with frequency; DC passes through.
VARIABLES X_L — inductive reactance, in ohms (Ω) f — frequency, in hertz (Hz) L — inductance, in henries (H)	UNITS & CONVERSION Convert L: $\text{mH} \times 10^{-3}$, $\mu\text{H} \times 10^{-6}$. Example: $10 \text{ }\mu\text{H}$ at $1 \text{ MHz} \rightarrow 6.283 \times 1\,000\,000 \times 0.000\,01 = 62.8 \text{ }\Omega$. Shortcut: $\text{MHz} \times \mu\text{H}$ works directly (the prefixes cancel): $6.283 \times 1 \times 10 = 62.8 \text{ }\Omega$.
Energy stored $E = \frac{1}{2} \times L \times I^2$	Energy held in an inductor's magnetic field while current flows.
VARIABLES E — energy, in joules (J) L — inductance, in henries (H) I — current through the inductor, in amps (A)	UNITS & CONVERSION L in H, I in A. Example: 100 mH at $2 \text{ A} \rightarrow 0.5 \times 0.1 \times 2^2 = 0.2 \text{ J}$.

RL time constant $\tau = L / R$	Time for the current in an inductor-resistor circuit to reach 63% of its final change.
VARIABLES τ — time constant, in seconds (s) L — inductance, in henries (H) R — total resistance in the loop, in ohms (Ω)	UNITS & CONVERSION $H \div \Omega = \text{seconds}$. Example: 10 mH with 100 $\Omega \rightarrow 0.01 \div 100 = 0.0001 \text{ s} = 100 \mu\text{s}$. Shortcut: mH $\div$ k Ω gives μs .
Induced voltage $V = L \times (\Delta I / \Delta t)$	Voltage produced when the current through an inductor changes. A fast change (switching off) causes a high spike, which is why relay coils need a flyback diode.
VARIABLES V — induced voltage, in volts L — inductance, in henries (H) ΔI — change in current, in amps Δt — time the change takes, in seconds	UNITS & CONVERSION Example: a 10 mH relay coil carrying 1 A is switched off in 1 $\mu\text{s} \rightarrow 0.01 \times (1 \div 0.000001) = 10000 \text{ V spike}$. Convert $\mu\text{s} \rightarrow \text{s} (\times 10^{-6})$ or the answer is a million times too small.

5. Resonance & Filters

Resonant frequency $f_0 = 1 / (2\pi\sqrt{LC})$	Frequency at which inductive and capacitive reactance are equal. This sets the frequency of LC tanks and oscillators.
VARIABLES f_0 — resonant frequency, in hertz (Hz) L — inductance, in henries (H) C — capacitance, in farads (F)	UNITS & CONVERSION L and C must be in H and F. Example: 10 μH and 100 pF $\rightarrow LC = 10^{-5} \times 10^{-10} = 10^{-15}$; $\sqrt{} = 3.16 \times 10^{-8}$; $f_0 = 1 / (6.283 \times 3.16 \times 10^{-8}) \approx 5.03 \text{ MHz}$. Shortcut: $f_0(\text{MHz}) \approx 159 / \sqrt{(L \text{ in } \mu\text{H} \times C \text{ in pF})} \rightarrow 159 / \sqrt{1000} = 5.03 \text{ MHz}$.
Q factor, series RLC $Q = (1/R) \times \sqrt{L/C}$	Sharpness of resonance in a series RLC circuit. Higher Q means a narrower frequency response.
VARIABLES Q — quality factor (no unit) R — total series resistance, including the coil's own wire resistance, in Ω L — inductance, in H C — capacitance, in F	UNITS & CONVERSION L and C in base units (H, F); units cancel so Q has no unit. Example: $R = 10 \Omega$, $L = 10 \mu\text{H}$, $C = 100 \text{ pF} \rightarrow \sqrt{(10^{-5}/10^{-10})} = \sqrt{10^5} = 316 \rightarrow Q = 31.6$.
Q factor, parallel RLC $Q = R \times \sqrt{C/L}$	Sharpness of resonance in a parallel RLC circuit. A higher parallel R gives a higher Q .
VARIABLES Q — quality factor (no unit) R — resistance in parallel with the tank, in Ω C — capacitance, in F L — inductance, in H	UNITS & CONVERSION C and L in F and H. Example: $R = 10 \text{ k}\Omega$, $L = 10 \mu\text{H}$, $C = 100 \text{ pF} \rightarrow \sqrt{(10^{-10}/10^{-5})} = 0.00316 \rightarrow Q = 10000 \times 0.00316 = 31.6$.
Bandwidth $BW = f_0 / Q$	Width of the frequency range between the -3 dB points around resonance.
VARIABLES BW — bandwidth, same unit as f_0 f_0 — resonant frequency Q — quality factor	UNITS & CONVERSION BW comes out in whatever unit f_0 is in. Example: $f_0 = 10 \text{ MHz}$, $Q = 50 \rightarrow BW = 0.2 \text{ MHz} = 200 \text{ kHz}$.

RC filter cutoff $f_c = 1 / (2\pi RC)$	Cutoff (−3 dB) frequency of an RC low-pass or high-pass filter.
VARIABLES f_c — cutoff frequency, in hertz (Hz) R — resistance, in ohms (Ω) C — capacitance, in farads (F)	UNITS & CONVERSION R in Ω , C in F. Example: 1 k Ω and 100 nF $\rightarrow 1 / (6.283 \times 1000 \times 10^{-7}) \approx 1592$ Hz. Shortcut: k Ω and μ F give kHz $\rightarrow 1 / (6.283 \times 1 \times 0.1) = 1.59$ kHz.
RL filter cutoff $f_c = R / (2\pi L)$	Cutoff (−3 dB) frequency of an RL low-pass or high-pass filter.
VARIABLES f_c — cutoff frequency, in hertz (Hz) R — resistance, in ohms (Ω) L — inductance, in henries (H)	UNITS & CONVERSION L in henries. Example: 100 Ω and 10 mH $\rightarrow 100 / (6.283 \times 0.01) \approx 1592$ Hz.

6. Diodes & LEDs

Diode forward drop (typical) Si: $V_f \approx 0.6\text{--}0.7$ V Schottky: $V_f \approx 0.2\text{--}0.4$ V Ge: $V_f \approx 0.3$ V	Voltage lost across a conducting diode. Silicon: general use (1N4148, 1N4007). Schottky: lower drop, faster (1N5819). Germanium: RF detectors.
VARIABLES V_f — forward voltage — the drop across the diode when current flows from anode to cathode, in volts	UNITS & CONVERSION V_f rises slightly with current and falls with temperature. Use the datasheet graph for the current you actually run.
LED forward voltage (typical) Red / yellow: 1.8–2.2 V Green: 2.0–3.2 V Blue / white: 3.0–3.4 V	Minimum voltage needed for the LED to conduct and light, depending on colour. Blue and white LEDs may not light from a 3.3 V pin with a series resistor.
VARIABLES V_f — LED forward voltage at its rated current (usually 10–20 mA), in volts	
LED series resistor $R = (V_{\text{supply}} - V_f) / I_f$	Resistor that limits LED current. The voltage left after the LED drop appears across the resistor.
VARIABLES R — resistor value, in ohms (Ω) V_{supply} — supply or pin voltage, in volts V_f — LED forward voltage, in volts I_f — desired LED current, in amps (A)	UNITS & CONVERSION Convert mA $\rightarrow$ A: 20 mA = 0.02 A. Example: 5 V supply, red LED 2 V, 20 mA $\rightarrow (5 - 2) / 0.02 = 150$ Ω . Or: 3 V $\div$ 20 mA = 0.15 k Ω . If the result isn't a standard value, round up (e.g. 150 $\rightarrow$ 150 Ω or 180 Ω).

7. BJT Transistors

Hand-calculation approximations — real h_{FE} varies a lot between parts, with current and with temperature.

DC current gain $I_C = h_{FE} \times I_B$	Collector current is the base current multiplied by the gain. Valid only in the active region, not in saturation.
VARIABLES I_C — collector current I_B — base current h_{FE} — DC current gain, no unit (datasheet, typically 100–300 for BC547 / 2N2222)	UNITS & CONVERSION I_C comes out in the same unit as I_B . Example: $I_B = 0.1$ mA, $h_{FE} = 200 \rightarrow I_C = 20$ mA.

Emitter current $I_E = I_B + I_C$	Emitter current is the sum of base and collector currents.
VARIABLES I_E — emitter current I_B — base current I_C — collector current	UNITS & CONVERSION Same unit for all three. Example: 0.1 mA + 20 mA = 20.1 mA.
Base resistor (switching) $R_B = (V_{in} - V_{BE}) / I_B$	Base resistor for using a BJT as a switch. For reliable saturation, use $I_B \approx I_C / 10$.
VARIABLES R_B — base resistor, in ohms (Ω) V_{in} — driving voltage, e.g. a 3.3 V or 5 V GPIO pin V_{BE} — base-emitter drop, ≈ 0.7 V I_B — base current, in amps (A)	UNITS & CONVERSION I_B in amps. Example: switch 100 mA from a 3.3 V pin $\rightarrow I_B = 10 \text{ mA} = 0.01 \text{ A} \rightarrow R_B = (3.3 - 0.7) / 0.01 = 260 \Omega \rightarrow$ use 270 Ω. Check the pin can supply that current.
Typical silicon BJT voltages $V_{BE}(\text{on}) \approx 0.6\text{-}0.7 \text{ V}$ $V_{CE}(\text{sat}) \approx 0.1\text{-}0.3 \text{ V}$	V_{BE}: needed to turn the transistor on. $V_{CE}(\text{sat})$: voltage remaining across a fully-on transistor.
VARIABLES V_{BE} — voltage between base and emitter, in volts $V_{CE}(\text{sat})$ — collector-emitter voltage when fully switched on, in volts	
Power dissipation $P_D = V_{CE} \times I_C$	Heat generated in the transistor. Must stay below the datasheet's $P_D(\text{max})$.
VARIABLES P_D — power dissipated, in watts (W) V_{CE} — voltage between collector and emitter, in volts I_C — collector current, in amps (A)	UNITS & CONVERSION I_C in amps for watts ($V \times \text{mA} = \text{mW}$). Example: saturated switch, $0.2 \text{ V} \times 0.5 \text{ A} = 0.1 \text{ W}$ — cool. Half-on at $6 \text{ V} \times 0.5 \text{ A} = 3 \text{ W}$ — needs a heatsink.

8. MOSFETs

$V_{GS}(\text{th})$, $R_{DS}(\text{on})$ and Q_g are always datasheet values for the specific part.

Gate threshold voltage $V_{GS}(\text{th})$	Gate voltage at which the MOSFET starts to conduct (typically at 250 μA). It is not the full turn-on voltage — use the V_{GS} at which $R_{DS}(\text{on})$ is specified.
VARIABLES V_{GS} — voltage between gate and source, in volts (th) — threshold — the turn-on starting point	
On-state power loss $P_D = I_D^2 \times R_{DS}(\text{on})$	Heat generated in a fully-on MOSFET due to its on-resistance.
VARIABLES P_D — power dissipated, in watts (W) I_D — drain current, in amps (A) $R_{DS}(\text{on})$ — on-resistance between drain and source, in ohms	UNITS & CONVERSION Datasheets list $R_{DS}(\text{on})$ in milliohms — divide by 1000: 25 mΩ = 0.025 Ω. Example: 5 A through 25 mΩ $\rightarrow 5^2 \times 0.025 = 0.625 \text{ W}$.
Gate drive current $I_G \approx Q_g / t_{sw}$	Current needed to charge the gate in the required switching time. Large MOSFETs need a gate driver, not a GPIO pin.
VARIABLES I_G — gate drive current, in amps (A) Q_g — total gate charge, in coulombs (datasheet gives nC) t_{sw} — desired switching time, in seconds	UNITS & CONVERSION Shortcut: nC $\div$ ns = A. Example: $Q_g = 40 \text{ nC}$ switched in 100 ns $\rightarrow 40 \div 100 = 0.4 \text{ A}$ peak.

Logic-level MOSFETs Fully on at $V_{GS} \approx 4.5 \text{ V}$ (or 2.5 V)	Standard MOSFETs (e.g. IRFZ44N) need $\approx 10 \text{ V}$ on the gate. Logic-level MOSFETs (e.g. IRLZ44N, AO3400) fully turn on at 4.5 V or less.
VARIABLES V_{GS} — gate-source voltage at which $R_{DS(on)}$ is guaranteed in the datasheet	UNITS & CONVERSION Check the $R_{DS(on)}$ line of the datasheet: it states “@ $V_{GS} = 10 \text{ V}$ ”, “@ 4.5 V” or “@ 2.5 V”. For 3.3 V drive you want a part rated at 2.5 V.

9. AC & Impedance

Impedance (complex) $Z = R + jX$	Total opposition to AC: resistance plus reactance.
VARIABLES Z — impedance, in ohms (Ω) R — resistive part, in ohms X — net reactance = $X_L - X_C$, in ohms (positive = inductive, negative = capacitive) j — the imaginary unit $\sqrt{-1}$ (engineers use j because i already means current)	UNITS & CONVERSION All parts in ohms — calculate X_L and X_C first with f in Hz, L in H, C in F.
Impedance magnitude $Z = \sqrt{R^2 + X^2}$	Single-value size of the impedance. Use it in Ohm's law for AC: $I = V / Z $.
VARIABLES $ Z $ — size of the impedance, in ohms R — resistance, in ohms X — net reactance, in ohms	UNITS & CONVERSION R and X in the same unit. Example: $R = 100 \Omega$, $X = 100 \Omega \rightarrow \sqrt{(10\,000 + 10\,000)} \approx 141 \Omega$. Then $I = V / Z $ works like Ohm's law.
Phase angle $\theta = \arctan(X / R)$	Time shift between voltage and current. Positive: current lags (inductive). Negative: current leads (capacitive).
VARIABLES θ — phase angle, in degrees or radians X — net reactance, in ohms R — resistance, in ohms	UNITS & CONVERSION X and R in the same unit. Set your calculator to DEG mode for degrees ($180^\circ = \pi \text{ rad}$). Example: $X = R \rightarrow \arctan(1) = 45^\circ$.
RMS from peak $V_{rms} = V_{peak} / \sqrt{2}$	RMS is the DC-equivalent value of an AC voltage. Multimeters (AC mode) and mains ratings show RMS.
VARIABLES V_{rms} — RMS voltage V_{peak} — peak voltage, measured from 0 V to the top of the wave $\sqrt{2} \approx 1.414$, so $1/\sqrt{2} \approx 0.707$	UNITS & CONVERSION Valid only for pure sine waves — square or distorted waves have different ratios. Example: 10 V peak $\rightarrow$ 7.07 V RMS.
Peak from RMS $V_{peak} = V_{rms} \times \sqrt{2}$	Peak value of a sine wave from its RMS value. A rectifier's smoothing capacitor charges to this value.
VARIABLES V_{peak} — peak voltage V_{rms} — RMS voltage	UNITS & CONVERSION Same unit in and out. Example: 230 V RMS $\times 1.414 \approx 325 \text{ V peak}$; 12 V AC transformer $\rightarrow \approx 17 \text{ V peak}$.

Peak-to-peak $V_{pp} = 2 \times V_{peak}$	Voltage from the lowest to the highest point of the waveform, as read on an oscilloscope.
VARIABLES V_{pp} — peak-to-peak voltage V_{peak} — peak voltage (half of V_{pp} for a symmetric wave)	UNITS & CONVERSION Example: 325 V peak $\rightarrow$ 650 V_{pp} . Don't confuse V_{pp} , V_{peak} and V_{rms} when reading specs.
Rectified average $V_{avg} = 0.637 \times V_{peak}$	Average value of a full-wave rectified sine wave before smoothing ($0.637 = 2/\pi$).
VARIABLES V_{avg} — average voltage V_{peak} — peak voltage	UNITS & CONVERSION Example: 17 V peak $\rightarrow$ 10.8 V average. Subtract diode drops (≈ 1.4 V for a bridge) from V_{peak} first for a real circuit.

10. 555 Timer

Astable frequency $f = 1.44 / ((R1 + 2 \times R2) \times C)$	Output frequency of a 555 in astable (free-running) mode.
VARIABLES f — output frequency, in hertz (Hz) $R1$ — resistor between V_{CC} and pin 7 (discharge), in ohms $R2$ — resistor between pin 7 and pins 2/6, in ohms C — timing capacitor from pins 2/6 to ground, in farads	UNITS & CONVERSION R in Ω and C in F give Hz. Example: $R1 = 1 \text{ k}\Omega$, $R2 = 10 \text{ k}\Omega$, $C = 10 \text{ }\mu\text{F} \rightarrow (1000 + 20\,000) \times 0.000\,01 = 0.21 \rightarrow f = 1.44 / 0.21 \approx 6.9 \text{ Hz}$. Shortcut: $\text{k}\Omega$ and μF give $\text{kHz} \rightarrow 1.44 / (21 \times 10) = 0.0069 \text{ kHz}$.
Astable duty cycle $D = (R1 + R2) / (R1 + 2 \times R2)$	Fraction of each cycle that the output is high. Always above 50% in this basic circuit.
VARIABLES D — duty cycle, as a fraction ($\times 100$ for %) $R1, R2$ — the timing resistors	UNITS & CONVERSION $R1$ and $R2$ only need the same unit. Example: $1 \text{ k}\Omega$ and $10 \text{ k}\Omega \rightarrow 11 / 21 = 0.52 = 52\%$.
Monostable pulse width $t = 1.1 \times R \times C$	Length of the single output pulse in monostable (one-shot) mode.
VARIABLES t — pulse length, in seconds R — timing resistor, in ohms C — timing capacitor, in farads	UNITS & CONVERSION $\Omega \times \text{F} = \text{s}$. Example: $100 \text{ k}\Omega$ and $10 \text{ }\mu\text{F} \rightarrow 1.1 \times 100\,000 \times 0.000\,01 = 1.1 \text{ s}$. Shortcut: $\text{k}\Omega \times \mu\text{F}$ gives $\text{ms} \rightarrow 1.1 \times 100 \times 10 = 1100 \text{ ms}$.

11. Decibels, Wavelength & Period

dB — power ratio $\text{dB} = 10 \times \log_{10}(P2 / P1)$	Power ratio in decibels. $+3 \text{ dB} \approx 2\times$, $+10 \text{ dB} = 10\times$, $-3 \text{ dB} = \frac{1}{2}$.
VARIABLES dB — gain (positive) or loss (negative), in decibels $P2$ — the power you are measuring (output) $P1$ — the reference power (input) $\log_{10}$ — base-10 logarithm, the “log” key	UNITS & CONVERSION $P1$ and $P2$ must be in the same unit (both W or both mW). Example: $1 \text{ W in, } 10 \text{ W out} \rightarrow 10 \times \log_{10}(10) = +10 \text{ dB}$.

dB — voltage / current ratio $dB = 20 \times \log_{10}(V2 / V1)$	Voltage or current ratio in decibels (same impedance). +6 dB $\approx$ 2$\times$, +20 dB = 10$\times$.
VARIABLES V2 — measured voltage (output) V1 — reference voltage (input)	UNITS & CONVERSION Same unit for both (both V or both mV). Example: 10 mV in, 1 V out $\rightarrow$ 1000 mV / 10 mV = 100 $\rightarrow$ 20 $\times$ 2 = +40 dB.
Wavelength $\lambda = c / f$	Physical length of one wave cycle at a given frequency. Used to size antennas.
VARIABLES λ — lambda, wavelength, in metres (m) c — speed of light, 3×10^8 m/s = 300 000 000 m/s f — frequency, in hertz (Hz)	UNITS & CONVERSION f in Hz for metres. Shortcut: λ (m) = 300 / f (MHz). Example: 100 MHz $\rightarrow$ 3 m; 433 MHz $\rightarrow$ 0.69 m.
Frequency and period $f = 1 / T$ $T = 1 / f$	Frequency and period are reciprocals of each other.
VARIABLES f — frequency, in hertz (Hz) T — period, in seconds (s)	UNITS & CONVERSION Seconds $\leftrightarrow$ Hz, milliseconds $\leftrightarrow$ kHz, microseconds $\leftrightarrow$ MHz. Example: T = 20 ms $\rightarrow$ f = 1/0.02 = 50 Hz (mains).

12. Wires, Circuits & General Laws

Wire resistance $R = \rho \times L / A$	Resistance of a conductor from its material, length and cross-section.
VARIABLES R — resistance, in ohms (Ω) ρ — rho, resistivity of the material: copper $1.68 \times 10^{-8} \Omega \cdot m$, aluminium $2.65 \times 10^{-8} \Omega \cdot m$ L — length of the conductor, in metres (m) A — cross-sectional area, in square metres (m^2)	UNITS & CONVERSION Convert $mm^2 \rightarrow m^2$ by $\times 10^{-6}$ ($1.5 mm^2 = 1.5 \times 10^{-6} m^2$). Easier: use $\rho = 0.0168 \Omega \cdot mm^2/m$ for copper, then L in m and A in mm^2 directly. Example: 20 m of 1.5 mm^2 copper $\rightarrow$ $0.0168 \times 20 / 1.5 = 0.224 \Omega$.
Voltage drop in a cable $V_{drop} = I \times R_{wire}$	Voltage lost in the cable between the source and the load.
VARIABLES V_{drop} — voltage lost in the cable, in volts I — load current, in amps (A) R_{wire} — resistance of both conductors — current goes out and back, so use 2 $\times$ cable length	UNITS & CONVERSION I in amps. Example: 10 m two-core 1.5 mm^2 cable at 10 A $\rightarrow$ loop is 20 m $\rightarrow$ $0.224 \Omega \rightarrow 10 \times 0.224 = 2.24$ V lost. On 12 V that's almost 19% — too much; on 230 V it's under 1%.
Charge $Q = I \times t$	Charge moved by a current in a given time.
VARIABLES Q — charge, in coulombs (C) I — current, in amps (A) t — time, in seconds (s)	UNITS & CONVERSION Battery link: 1 mAh = 0.001 A $\times$ 3600 s = 3.6 C. Example: 2 A for 10 s = 20 C.
Kirchhoff's Voltage Law $\Sigma V = 0$ around any closed loop	The sum of all voltages around any closed loop is zero.
VARIABLES Σ — sigma, "sum of" V — each voltage rise or drop in the loop	UNITS & CONVERSION Pick a direction, count rises as + and drops as -. Example: 9 V battery, LED drops 2 V, resistor 7 V $\rightarrow$ $9 - 2 - 7 = 0$.

Kirchhoff's Current Law $\Sigma I_{in} = \Sigma I_{out}$		The sum of currents entering a node equals the sum of currents leaving it.
VARIABLES ΣI_{in} — sum of currents entering the node ΣI_{out} — sum of currents leaving the node		UNITS & CONVERSION Same unit for all. Example: 50 mA in → 30 mA + 20 mA out.
Power factor $PF = P_{real} / P_{apparent}$		Ratio of real power to apparent power in an AC circuit (1 = ideal). Relevant for motors, transformers and switching supplies.
VARIABLES PF — power factor, 0 to 1, no unit P_{real} — real power, in watts (W) — what a wattmeter reads $P_{apparent}$ — apparent power = $V_{rms} \times I_{rms}$, in volt-amperes (VA)		UNITS & CONVERSION VA and W are the same size unit but mean different things. Example: 230 V × 1 A = 230 VA; wattmeter reads 184 W → PF = 0.8. Generators and UPSs are often rated in VA.

13. Microcontrollers & Digital Circuits

Written with typical 3.3 V / 5 V boards in mind (Arduino, ESP32, STM32, RP2040).

ADC reading $Reading = V_{in} / V_{ref} \times (2^n - 1)$		Raw ADC value for a given input voltage. Reverse: $V_{in} = Reading \times V_{ref} / (2^n - 1)$.
VARIABLES $Reading$ — raw ADC value (integer) V_{in} — voltage on the ADC pin, in volts V_{ref} — ADC reference / full-scale voltage, in volts (often 3.3 V or 5 V) n — ADC resolution in bits (10-bit → 0–1023, 12-bit → 0–4095)		UNITS & CONVERSION V_{in} and V_{ref} in the same unit. Example: 1.65 V on a 12-bit, 3.3 V ADC → $0.5 \times 4095 \approx 2047$. Note: the ESP32 ADC is noticeably non-linear near its ends.
ADC step size $V_{step} = V_{ref} / 2^n$		Smallest voltage change the ADC can detect.
VARIABLES V_{step} — resolution per count, in volts V_{ref} — reference voltage, in volts n — resolution in bits		UNITS & CONVERSION Answer is in volts — × 1000 for mV. Example: 10-bit at 3.3 V → $3.3 / 1024 = 0.00322 \text{ V} = 3.22 \text{ mV}$.
Nyquist sampling rate $f_s \geq 2 \times f_{max}$		Minimum sampling rate that avoids aliasing. In practice, sample 5–10× the highest frequency.
VARIABLES f_s — sampling rate, in samples per second (Hz) f_{max} — highest frequency in the signal, in Hz		UNITS & CONVERSION Same unit both sides. Example: audio up to 20 kHz → at least 40 kHz (CDs use 44.1 kHz).
PWM $f = 1/T$ $D = t_{on} / T$ $V_{avg} = D \times V_{high}$		Frequency, duty cycle and average voltage of a PWM signal.
VARIABLES f — PWM frequency, in Hz T — period of one on+off cycle, in seconds t_{on} — time the output is high in each cycle D — duty cycle, fraction (× 100 for %) V_{avg} — average output voltage V_{high} — the pin's high voltage (3.3 V / 5 V)		UNITS & CONVERSION t_{on} and T in the same unit. Example: $T = 1 \text{ ms}$ (1 kHz), $t_{on} = 0.25 \text{ ms}$ → $D = 25\%$ → $V_{avg} = 0.25 \times 3.3 = 0.83 \text{ V}$. 8-bit PWM value 64 → $D = 64/255 \approx 25\%$.

GPIO LED resistor $R = (V_{\text{gpio}} - V_f) / I_{\text{led}}$		Series resistor for an LED driven directly from a GPIO pin. Keep the current within the pin's datasheet limit.
VARIABLES R — resistor, in ohms V_{gpio} — pin high voltage, in volts V_f — LED forward voltage I_{led} — LED current, in amps		UNITS & CONVERSION mA → A. Example: 3.3 V pin, red LED 2 V, 10 mA → (3.3 – 2) / 0.01 = 130 Ω → use 150 Ω.
Decoupling capacitors ≈ 100 nF at every IC power pin + 10-100 μF bulk per board		Standard practice: a 100 nF ceramic capacitor next to every IC power pin, plus a larger bulk capacitor near the power input.
VARIABLES 100 nF — = 0.1 μF = ceramic marked “104” bulk — an electrolytic or large ceramic near the power input or regulator		UNITS & CONVERSION 100 nF = 0.1 μF = 100 000 pF — all the same part. Place it within a few mm of the pin, with a short path to ground.
Crystal load capacitors $C_L = (C1 \times C2) / (C1 + C2) + C_{\text{stray}}$ $C1 = C2 = 2 \times (C_L - C_{\text{stray}})$		First formula: total load capacitance seen by the crystal. Second: the value of C1 and C2 to fit for the C _L given in the datasheet.
VARIABLES C_L — load capacitance from the crystal datasheet C1, C2 — the two capacitors from each crystal pin to ground C_{stray} — PCB and chip-pin capacitance, typically 2-5 pF		UNITS & CONVERSION Everything in pF. Example: C _L = 18 pF, C _{stray} = 3 pF → C1 = C2 = 2 × 15 = 30 pF → use 27 or 33 pF.
UART bit time $t_{\text{bit}} = 1 / \text{baud}$		Duration of one bit. A byte in 8N1 format takes 10 bits.
VARIABLES t_{bit} — duration of one bit, in seconds baud — baud rate, in bits per second		UNITS & CONVERSION Answer in seconds — × 10 ⁶ for μs. Example: 115 200 baud → 8.68 μs per bit; one byte ≈ 86.8 μs → max ≈ 11 520 bytes/s.
I ² C pull-up — minimum $R_{\text{min}} = (V_{\text{DD}} - V_{\text{OL}}) / I_{\text{OL}}$		Smallest allowed I ² C pull-up resistor.
VARIABLES V_{DD} — bus supply voltage, in volts V_{OL} — maximum “low” level, 0.4 V in the I ² C spec I_{OL} — sink current a device must handle, 3 mA (standard / fast mode)		UNITS & CONVERSION I _{OL} in amps. Example: 3.3 V bus → (3.3 – 0.4) / 0.003 ≈ 967 Ω minimum.
I ² C pull-up — maximum $R_{\text{max}} = t_r / (0.8473 \times C_b)$		Largest allowed I ² C pull-up resistor for the bus speed. Choose a value between R _{min} and R _{max} ; typical: 4.7 kΩ (100 kHz), 2.2 kΩ (400 kHz).
VARIABLES t_r — maximum rise time: 1000 ns (100 kHz) or 300 ns (400 kHz) C_b — total bus capacitance (wires + every device's pins)		UNITS & CONVERSION ns → s (× 10 ⁻⁹), pF → F (× 10 ⁻¹²). Example: 400 kHz, 200 pF → 300×10 ⁻⁹ / (0.8473 × 200×10 ⁻¹²) ≈ 1770 Ω.

14. PCB Layout & Signal Integrity

Approximate formulas — for real boards, confirm with your PCB tool's calculator (KiCad has one built in).

Trace current capacity (IPC-2221) $I = k \times \Delta T^{0.44} \times A^{0.725}$	Current a trace can carry for a given temperature rise. Usually solved for the required trace width.
VARIABLES I — current, in amps (A) k — 0.048 for outer layers, 0.024 for inner layers ΔT — allowed temperature rise, in °C (10 °C is typical) A — trace cross-section, in square mils (mil ² = width × thickness)	UNITS & CONVERSION This one uses imperial units: 1 mil = 0.0254 mm; 1 oz copper = 1.37 mil (35 µm) thick. Example: 1 A, 10 °C, outer 1 oz → A ≈ 16 mil ² → width = 16 / 1.37 ≈ 12 mil ≈ 0.30 mm.
Trace resistance $R = \rho \times L / (W \times t)$	Resistance of a PCB trace.
VARIABLES R — resistance, in ohms ρ — copper resistivity 1.68×10 ⁻⁸ Ω·m L — trace length, in m W — trace width, in m t — copper thickness, in m (1 oz = 35 µm)	UNITS & CONVERSION Convert everything to metres: mm × 10 ⁻³ , µm × 10 ⁻⁶ . Example: 100 mm × 0.25 mm × 35 µm → 1.68×10 ⁻⁸ × 0.1 / (0.25×10 ⁻³ × 35×10 ⁻⁶) ≈ 0.19 Ω.
Microstrip impedance (IPC-2141) $Z_0 \approx 87 / \sqrt{(\epsilon_r + 1.41)} \times \ln(5.98h / (0.8w + t))$	Characteristic impedance of an outer-layer trace above a ground plane. Required for USB, Ethernet and RF lines (typically 50 Ω, or 90 Ω differential).
VARIABLES Z₀ — characteristic impedance, in ohms ϵ_r — relative permittivity of the board, ≈ 4.2–4.6 for FR4 h — dielectric thickness from trace to ground plane w — trace width t — copper thickness ln — natural logarithm	UNITS & CONVERSION h, w and t just need the <i>same</i> unit (all mm or all mil) — they appear as a ratio. Roughly accurate for w/h between 0.1 and 2. Example: 1.6 mm 2-layer board, w = 3 mm, t = 0.035 mm, ϵ_r = 4.3 → ≈ 50 Ω.
Propagation delay (microstrip) $t_{pd} = 85 \times \sqrt{(0.475 \times \epsilon_r + 0.67)}$	Signal travel time per inch of microstrip trace.
VARIABLES t_{pd} — delay, in picoseconds per inch ϵ_r — board permittivity (FR4 ≈ 4.3)	UNITS & CONVERSION Result is ps/inch — divide by 25.4 for ps/mm. Example: ϵ_r = 4.3 → ≈ 140 ps/inch ≈ 5.5 ps/mm.
Critical trace length $L_{crit} \approx t_{rise} / (6 \times t_{pd})$	Trace length above which the trace must be treated as a transmission line (controlled impedance, termination).
VARIABLES L_{crit} — critical length t_{rise} — signal rise time (datasheet) t_{pd} — propagation delay per unit length	UNITS & CONVERSION t _{rise} and t _{pd} in the same time unit (ps); the length unit comes from t _{pd} . Example: 1 ns = 1000 ps rise, 140 ps/inch → 1000 / 840 ≈ 1.2 inch ≈ 30 mm.
Skin depth $\delta = \sqrt{(\rho / (\pi \times f \times \mu))}$	Depth of the outer layer of a conductor that carries current at a given frequency.
VARIABLES δ — skin depth, in metres ρ — resistivity, copper 1.68×10 ⁻⁸ Ω·m f — frequency, in Hz μ — permeability = 4π×10 ⁻⁷ H/m for copper	UNITS & CONVERSION f in Hz. Copper shortcut: δ (µm) ≈ 66 / √f (MHz). Example: 1 MHz → 66 µm; 100 MHz → 6.6 µm (thinner than 1 oz copper).

Capacitor self-resonance $f_{\text{SRF}} = 1 / (2\pi\sqrt{\text{ESL} \times \text{C}})$	Frequency above which a capacitor acts as an inductor due to its package and trace inductance.
VARIABLES f_{SRF} — self-resonant frequency, in Hz ESL — equivalent series inductance, in H ($\approx 0.5\text{--}1$ nH for 0402/0603, more with long traces) C — capacitance, in F	UNITS & CONVERSION nH $\times 10^{-9}$, nF $\times 10^{-9}$. Example: 100 nF with 1 nH $\rightarrow 1 / (6.283 \times \sqrt{(10^{-9} \times 10^{-7})}) = 1 / (6.283 \times 10^{-8}) \approx 15.9$ MHz.

15. Batteries & Power Storage

Capacity $\text{Ah} = \text{I} \times \text{t}$	Charge a battery delivers: current multiplied by time in hours.
VARIABLES Ah — capacity, in amp-hours I — current, in amps t — time, in hours (h)	UNITS & CONVERSION Time here is in hours , not seconds. mAh $\div 1000 = \text{Ah}$ (2500 mAh = 2.5 Ah).
C-rate $\text{C-rate} = \text{I} / \text{capacity}$	Charge or discharge current relative to battery capacity. 1C $\approx$ full discharge in 1 hour.
VARIABLES I — charge or discharge current capacity — battery capacity	UNITS & CONVERSION Use matching units: A with Ah, or mA with mAh. Example: 1000 mA from a 2000 mAh cell = 0.5C.
Runtime estimate $\text{t} \approx \text{capacity} / \text{I}_{\text{load}}$	Approximate runtime at constant load. Real runtime is typically 70–90% of this value.
VARIABLES t — runtime, in hours capacity — battery capacity I_{load} — average load current	UNITS & CONVERSION mAh with mA, or Ah with A. Example: 2000 mAh $\div 80$ mA = 25 h $\rightarrow$ expect ≈ 20 h.
Energy stored $\text{Wh} = \text{Ah} \times \text{V}$	Total energy stored. Used to compare batteries of different voltages.
VARIABLES Wh — energy, in watt-hours Ah — capacity, in amp-hours V — nominal voltage, in volts	UNITS & CONVERSION Ah, not mAh. Example: 3.7 V 2500 mAh = $2.5 \times 3.7 = 9.25$ Wh; 12 V 7 Ah = 84 Wh. (1 Wh = 3600 J.)
Voltage sag under load $\text{V}_{\text{terminal}} = \text{V}_{\text{oc}} - \text{I} \times \text{R}_{\text{int}}$	Terminal voltage drop under load caused by internal resistance.
VARIABLES $\text{V}_{\text{terminal}}$ — voltage at the terminals under load V_{oc} — open-circuit (no-load) voltage I — load current, in amps R_{int} — internal resistance, in ohms	UNITS & CONVERSION R_{int} is quoted in milliohms: 50 m Ω = 0.05 Ω . Example: 4.0 V cell, 50 m Ω , 2 A $\rightarrow 4.0 - 0.1 = 3.9$ V.
State of charge Read from a chemistry-specific curve	Estimated from resting (no-load) voltage using the chemistry's discharge curve. There is no single formula.
VARIABLES Rough Li-ion values — 4.2 V $\approx$ full, 3.7 V $\approx$ half, 3.0 V $\approx$ empty (don't go lower)	

16. RF & Antennas

Relevant to Colpitts oscillators, SDR reception and antenna experiments.

Quarter-wave antenna $L = c / (4 \times f)$	Length of a quarter-wave antenna (whip or ground-plane).
VARIABLES L — antenna length, in metres c — speed of light, 3×10^8 m/s f — frequency, in Hz	UNITS & CONVERSION f in Hz → metres. Shortcut: $L \text{ (cm)} \approx 7500 / f \text{ (MHz)}$. Example: 433 MHz → 17.3 cm. Real antennas are cut about 5% shorter (end effect).
Half-wave dipole $L = c / (2 \times f)$	Total length of a half-wave dipole; each arm is a quarter wave.
VARIABLES L — total dipole length, in metres c — speed of light f — frequency, in Hz	UNITS & CONVERSION Shortcut: $L \text{ (m)} \approx 150 / f \text{ (MHz)}$, practical $\approx 143 / f \text{ (MHz)}$. Example: 100 MHz → about 1.43 m total, 0.71 m per arm.
Free-space path loss $FSPL = 20 \cdot \log_{10}(d) + 20 \cdot \log_{10}(f) + 32.44$	Signal loss over a distance in free space, without obstacles.
VARIABLES FSPL — path loss, in dB d — distance, in kilometres f — frequency, in megahertz	UNITS & CONVERSION The constant 32.44 only works with km and MHz. With d in metres use -27.55 instead. Example: 1 km at 433 MHz → $0 + 52.7 + 32.4 \approx 85$ dB.
Reflection coefficient $\Gamma = (Z_L - Z_0) / (Z_L + Z_0)$	Fraction of the signal reflected at an impedance mismatch. 0 = matched, 1 = full reflection.
VARIABLES Γ — gamma, reflection coefficient, 0-1 Z_L — load impedance (antenna), in ohms Z_0 — line impedance (coax, usually 50 Ω), in ohms	UNITS & CONVERSION Both in ohms. Example: 75 Ω antenna on 50 Ω coax → $(75 - 50) / (75 + 50) = 0.2$.
VSWR $VSWR = (1 + \Gamma) / (1 - \Gamma)$	Standing wave ratio. 1:1 = perfect match; below 2:1 is usually acceptable.
VARIABLES VSWR — voltage standing wave ratio (written as X:1) $ \Gamma $ — size of the reflection coefficient	UNITS & CONVERSION No units. Example: $\Gamma = 0.2 \rightarrow 1.2 / 0.8 = 1.5:1$.
Return loss $RL = -20 \times \log_{10} \Gamma$	Reflected power expressed in dB. Higher = better match.
VARIABLES RL — return loss, in dB $ \Gamma $ — size of the reflection coefficient	UNITS & CONVERSION Example: $\Gamma = 0.2 \rightarrow -20 \times \log_{10}(0.2) = 14$ dB.
dBm ↔ milliwatts $P = 10^{dBm/10}$ $dBm = 10 \times \log_{10}(P)$	Conversion between dBm (power relative to 1 mW) and milliwatts.
VARIABLES P — power, in milliwatts (mW) dBm — power in decibels relative to 1 mW	UNITS & CONVERSION P must be in mW (1 W = 1000 mW). Examples: 0 dBm = 1 mW, 10 dBm = 10 mW, 20 dBm = 100 mW, 30 dBm = 1 W, -100 dBm = 0.1 pW.

Far-field distance $r \approx 2D^2 / \lambda$	Distance beyond which an antenna is in the far field, where standard radiation formulas apply.
VARIABLES r — far-field boundary distance D — largest dimension of the antenna λ — wavelength	UNITS & CONVERSION D and λ in the same unit; r comes out in that unit. Example: D = 1 m, λ = 0.3 m (1 GHz) → $2 \times 1 / 0.3 \approx 6.7$ m.

Quick reference only — always confirm device-specific limits ($V_{CE(max)}$, $R_{DS(on)}$, $P_D(max)$, pin currents) against the actual datasheet. PCB and RF formulas are simplified approximations — use proper calculators and simulators for real designs.

17. Solving Circuits — Finding Unknown Values

Every circuit problem uses three tools: **Ohm's law** for each component, plus **Kirchhoff's current law** (KCL) and **voltage law** (KVL) for how components connect. The table shows what each connection type shares and what adds up.

	Series — one path, end to end	Parallel — across the same two points
Current	Same through every component	Splits between branches: $I = I_1 + I_2 + \dots$ (KCL)
Voltage	Splits between components: $V = V_1 + V_2 + \dots$ (KVL)	Same across every branch
Resistance	Adds: $R_t = R_1 + R_2 + \dots$	Reciprocals add: $1/R_t = 1/R_1 + 1/R_2 + \dots$
Total vs parts	Total is larger than any part	Total is smaller than any branch

General method for any unknown component

Step	What to do	Why
1	Draw the circuit. Mark every known value and the unknown.	Makes series and parallel groups visible.
2	Total resistance seen by the source: $R_t = V / I_{\text{total}}$	Ohm's law applied to the whole circuit.
3	Work from the source toward the unknown. For each known part find its voltage (series) or current (parallel).	Series parts carry the full current; parallel parts see the full voltage.
4	Subtract: remaining voltage (KVL) or remaining current (KCL) belongs to the unknown part.	What isn't used by known parts must go through the unknown.
5	Unknown $R = V$ across it $\div I$ through it.	Ohm's law applied to the unknown alone.
6	Check: put the result back and recompute R_t or I_{total}	Catches unit and arithmetic mistakes.

Two equivalent routes: **Method A** follows currents and voltages branch by branch (easy to check with a multimeter).

Method B reduces the circuit to one equivalent resistance and then removes the known parts from it. Both give the same answer.

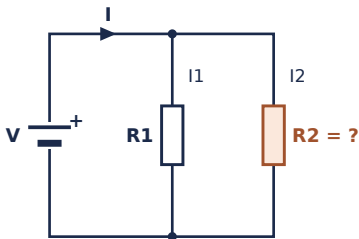
Reverse formulas — remove a known component from a measured total

Component	Unknown in series with known X1	Unknown in parallel with known X1
Resistor	$R_x = R_t - R_1$	$R_x = (R_1 \times R_t) / (R_1 - R_t)$
Capacitor	$C_x = (C_1 \times C_t) / (C_1 - C_t)$	$C_x = C_t - C_1$
Inductor	$L_x = L_t - L_1$	$L_x = (L_1 \times L_t) / (L_1 - L_t)$

The product-over-difference form comes from $1/X_x = 1/X_t - 1/X_1$. It only works if the total is smaller than the known part (for parallel R and L, series C). A negative or infinite result means the measurements are wrong. Units: same prefix for all values — the answer keeps it.

17.1 Resistor networks

Problem 1 — Unknown resistor in parallel



GIVEN supply voltage V , total current I , resistor $R1$.

FIND $R2$, connected in parallel with $R1$.

METHOD A — branch currents and voltages

1 $I1 = V / R1$

Parallel: $R1$ has the full supply voltage across it.

2 $I2 = I - I1$

KCL: total current splits; the rest flows through $R2$.

3 $R2 = V / I2$

$R2$ also has the full voltage across it.

METHOD B — equivalent resistance

1 $R_t = V / I$

Resistance of the whole parallel pair.

2 $R2 = R1 \times R_t / (R1 - R_t)$

Removes $R1$ from the parallel total.

WORKED EXAMPLE

$V = 12\text{ V}$, $I = 18\text{ mA}$, $R1 = 1\text{ k}\Omega$ (mA and k Ω work together directly).
 $I1 = 12\text{ V} / 1\text{ k}\Omega = 12\text{ mA} \rightarrow I2 = 18 - 12 = 6\text{ mA} \rightarrow R2 = 12\text{ V} / 6\text{ mA} = \mathbf{2\text{ k}\Omega}$.
 Check: $1\text{ k}\Omega \parallel 2\text{ k}\Omega = 0.667\text{ k}\Omega$; $12\text{ V} / 0.667\text{ k}\Omega = 18\text{ mA} \checkmark$

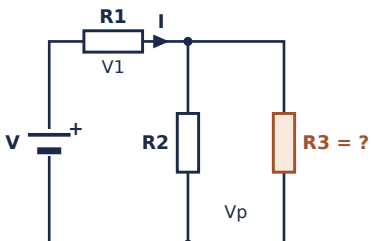
DIRECT FORMULA

$$R_t = V / I$$

$$R2 = (R1 \times R_t) / (R1 - R_t)$$

Requires $I > V / R1$. If not, $R2$ would be negative — recheck the measurements.

Problem 2 — Unknown resistor in a series-parallel circuit



GIVEN supply voltage V , total current I , $R1$ (series), $R2$ (parallel branch).

FIND $R3$, connected in parallel with $R2$; the pair is in series with $R1$.

METHOD A — branch currents and voltages

1 $V1 = I \times R1$

All current passes through $R1$ (series).

2 $Vp = V - V1$

KVL: the remaining voltage is across the parallel pair.

3 $I2 = Vp / R2$

Current in the known branch.

4 $I3 = I - I2$

KCL: the rest flows through $R3$.

5 $R3 = Vp / I3$

Ohm's law for $R3$ alone.

METHOD B — equivalent resistance

1 $R_t = V / I$

Total resistance of the circuit.

2 $R_p = R_t - R1$

Remove the series part: what's left is $R2 \parallel R3$.

3 $R3 = R2 \times R_p / (R2 - R_p)$

Remove $R2$ from the parallel pair.

WORKED EXAMPLE

$V = 12\text{ V}$, $I = 6\text{ mA}$, $R1 = 1\text{ k}\Omega$, $R2 = 2\text{ k}\Omega$.
 $V1 = 6\text{ mA} \times 1\text{ k}\Omega = 6\text{ V} \rightarrow Vp = 12 - 6 = 6\text{ V} \rightarrow I2 = 6\text{ V} / 2\text{ k}\Omega = 3\text{ mA}$
 $\rightarrow I3 = 6 - 3 = 3\text{ mA} \rightarrow R3 = 6\text{ V} / 3\text{ mA} = \mathbf{2\text{ k}\Omega}$.
 Check: $2\text{ k}\Omega \parallel 2\text{ k}\Omega = 1\text{ k}\Omega$; $+ R1 = 2\text{ k}\Omega$; $12\text{ V} / 2\text{ k}\Omega = 6\text{ mA} \checkmark$

DIRECT FORMULA

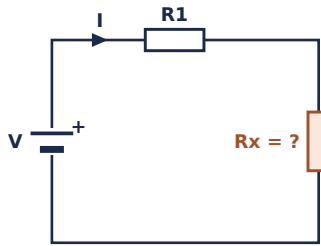
$$R_t = V / I$$

$$R_p = R_t - R1$$

$$R3 = (R2 \times R_p) / (R2 - R_p)$$

Requires $R_p < R2$.

Problem 3 — Unknown resistor in series



GIVEN supply voltage V , current I (or the voltage V_1 across R_1), resistor R_1 .
FIND R_x , in series with R_1 .

1 $I = V_1 / R_1$

Only needed if you measured V_1 instead of I .

2 $V_x = V - I \times R_1$

KVL: supply voltage minus the drop on R_1 .

3 $R_x = V_x / I$

Series: same current through R_x .

WORKED EXAMPLE

$V = 9\text{ V}$, $I = 15\text{ mA}$, $R_1 = 330\ \Omega \rightarrow V_1 = 0.015 \times 330 = 4.95\text{ V} \rightarrow V_x = 4.05\text{ V} \rightarrow R_x = 4.05 / 0.015 = \mathbf{270\ \Omega}$.

Only voltages known? $V = 9\text{ V}$, $V_1 = 4.95\text{ V}$: $I = 4.95 / 330 = 15\text{ mA}$, then as above. At least one resistance must be known — two voltages alone are not enough.

DIRECT FORMULA

$R_x = V / I - R_1$

or, from voltages:

$R_x = R_1 \times (V - V_1) / V_1$

Problem 4 — Any two-terminal circuit as one source (Thévenin)

GIVEN a circuit feeding a load — e.g. a divider, sensor output, or battery with wiring.

FIND load voltage and current for any load R_L , without re-solving the whole circuit.

1 $V_{th} = \text{open-circuit voltage}$

Measure or calculate the output with no load.

2 $R_{th} = V_{oc} / I_{sc}$

Or: turn sources off (replace with a wire) and find the resistance seen from the output.

3 $I_L = V_{th} / (R_{th} + R_L)$

The circuit behaves like one source V_{th} with R_{th} in series.

4 $V_L = I_L \times R_L$

Voltage the load actually gets.

WORKED EXAMPLE

Divider 12 V , $R_1 = R_2 = 10\text{ k}\Omega$: $V_{th} = 6\text{ V}$, $R_{th} = 10\text{ k}\Omega \parallel 10\text{ k}\Omega = 5\text{ k}\Omega$.
 Load $10\text{ k}\Omega \rightarrow I_L = 6\text{ V} / 15\text{ k}\Omega = 0.4\text{ mA} \rightarrow V_L = \mathbf{4\text{ V}}$ (not 6 V — the load pulls the divider down).

Rule: a divider output stays within $\sim 10\%$ if $R_L \geq 10 \times R_{th}$.

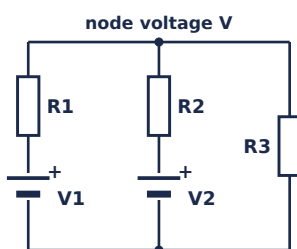
DIRECT FORMULA

$V_L = V_{th} \times R_L / (R_{th} + R_L)$

For a divider: $V_{th} = V_{in} \times R_2 / (R_1 + R_2)$, $R_{th} = R_1 \parallel R_2$.

Maximum power to the load when $R_L = R_{th}$: $P = V_{th}^2 / (4R_{th})$.

Problem 5 — Two sources feeding one node (Millman)



GIVEN sources V_1 , V_2 with series resistors R_1 , R_2 , and a load R_3 — all joined at one node.

FIND the node voltage V and each branch current.

1 $V = (V_1/R_1 + V_2/R_2) / (1/R_1 + 1/R_2 + 1/R_3)$

Weighted average of the sources. A branch without a source adds only $1/R$ to the bottom.

2 $I_1 = (V_1 - V) / R_1$

Current from each source into the node.

3 $I_2 = (V_2 - V) / R_2$

Negative result = current flows back into that source.

4 $I_3 = V / R_3$

Check: $I_1 + I_2 = I_3$ (KCL).

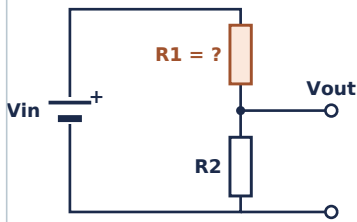
WORKED EXAMPLE

$V_1 = 12\text{ V}$, $V_2 = 5\text{ V}$, $R_1 = R_2 = R_3 = 100\ \Omega \rightarrow V = (0.12 + 0.05) / 0.03 = \mathbf{5.67\text{ V}}$.

$I_1 = 63.3\text{ mA}$, $I_2 = -6.7\text{ mA}$ (the 12 V source back-feeds the 5 V one), $I_3 = 56.7\text{ mA}$ ✓. This is why two supplies are joined through diodes (diode-OR).

17.2 Voltage dividers, sensors and bridges

Problem 6 — Design a divider for an ADC input



GIVEN maximum input voltage $V_{in,max}$, maximum safe ADC voltage V_{out} , a chosen $R2$.

FIND $R1$, and the formula to convert the ADC reading back to the input voltage.

1	$R1 = R2 \times (V_{in,max} - V_{out}) / V_{out}$	From $V_{out} = V_{in} \times R2 / (R1 + R2)$.
2	Round $R1$ up to a standard value	A larger $R1$ lowers V_{out} — safer for the pin.
3	$V_{in} = V_{adc} \times (R1 + R2) / R2$	Converts the measured pin voltage back to the input.
4	$I = V_{in} / (R1 + R2)$	Current the divider draws continuously.

WORKED EXAMPLE

Measure up to 15 V on a 3.3 V ADC, target 3.0 V, $R2 = 10\text{ k}\Omega \rightarrow R1 = 10\text{ k}\Omega \times 12 / 3 = 40\text{ k}\Omega \rightarrow \mathbf{39\text{ k}\Omega}$ (at 15 V: $15 \times 10 / 49 = 3.06\text{ V}$ ✓).
Readback: $V_{in} = V_{adc} \times 4.9$. Divider current at 12 V: 0.24 mA.
5 V $\rightarrow$ 3.3 V logic level shift: $R1 = 1\text{ k}\Omega$, $R2 = 2\text{ k}\Omega \rightarrow 3.33\text{ V}$.

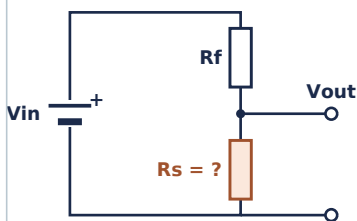
DIRECT FORMULA

$$R1 = R2 \times (V_{in} / V_{out} - 1)$$

$$R2 = R1 \times V_{out} / (V_{in} - V_{out})$$

Keep $R1 + R2 \leq \sim 100\text{ k}\Omega$ for most MCU ADCs; higher values give noisy readings.

Problem 7 — Unknown resistance from a divider reading (NTC, LDR, potentiometer)



GIVEN supply V_{in} , known fixed resistor R_f (top), measured V_{out} across the sensor (bottom).

FIND sensor resistance R_s .

1	$I = (V_{in} - V_{out}) / R_f$	Current through the fixed resistor (voltage across it $\div$ its value).
2	$R_s = V_{out} / I$	Series: the same current flows through the sensor.
3	$V_{out} = \text{reading} \times V_{ref} / (2^n - 1)$	When V_{out} comes from an ADC.

WORKED EXAMPLE

3.3 V, $R_f = 10\text{ k}\Omega$, 12-bit ADC reads 1365 $\rightarrow V_{out} = 1365 \times 3.3 / 4095 = 1.10\text{ V}$.
 $I = (3.3 - 1.1) / 10\text{ k}\Omega = 0.22\text{ mA} \rightarrow R_s = 1.10\text{ V} / 0.22\text{ mA} = \mathbf{5\text{ k}\Omega}$. This also answers “two voltages known”: it works because R_f is known.

DIRECT FORMULA

Sensor at bottom:

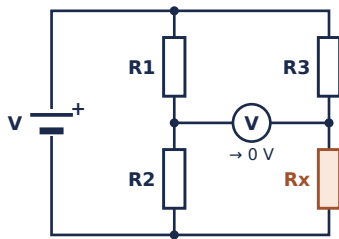
$$R_s = R_f \times V_{out} / (V_{in} - V_{out})$$

Sensor at top:

$$R_s = R_f \times (V_{in} - V_{out}) / V_{out}$$

Best resolution when $R_f \approx$ sensor value mid-range.

Problem 8 — Wheatstone bridge (precise unknown resistance)



GIVEN R1 and R2 (left branch), adjustable R3 (right top), meter between the two midpoints.

FIND Rx (right bottom).

1 Adjust R3 until the meter reads 0 V Balance: both midpoints are at the same voltage.

2 $R1 / R2 = R3 / Rx$ At balance both branches divide the supply in the same ratio.

3 $Rx = R2 \times R3 / R1$ Rearranged. The supply voltage cancels out.

WORKED EXAMPLE

R1 = 1 kΩ, R2 = 10 kΩ, balance at R3 = 470 Ω → Rx = 10 kΩ × 470 Ω / 1 kΩ = **4.7 kΩ**.

Used for strain gauges, load cells (HX711 modules) and precise resistance measurement. Accuracy depends on R1, R2, R3 — not on the meter or supply.

DIRECT FORMULA

$Rx = R2 \times R3 / R1$

Unbalanced output:

$$V_{out} = V \times (R2/(R1+R2) - Rx/(R3+Rx))$$

17.3 Capacitors, inductors and timing

Problem 9 — Time for a capacitor to reach a voltage

GIVEN supply V_s , R , C , threshold voltage V (e.g. MCU reset or logic threshold).

FIND time t — or the R (or C) needed for a target delay.

1	$\tau = R \times C$	Time constant in seconds ($k\Omega \times \mu F = ms$).
2	Charging: $t = \tau \times \ln(V_s / (V_s - V))$	Solved from $V = V_s (1 - e^{-t/\tau})$.
3	Discharging: $t = \tau \times \ln(V_0 / V)$	Solved from $V = V_0 \times e^{-t/\tau}$.
4	$R = t / (C \times \ln(V_s / (V_s - V)))$	Same equation solved for R (swap R and C to find C).

WORKED EXAMPLE

Power-on delay: 3.3 V, $R = 10\text{ k}\Omega$, $C = 10\text{ }\mu F$, threshold 2.0 V $\rightarrow \tau = 100\text{ ms} \rightarrow t = 100\text{ ms} \times \ln(3.3/1.3) = 100 \times 0.93 = \mathbf{93\text{ ms}}$.
For 500 ms: $R = 0.5 / (10\text{ }\mu F \times 0.93) = 53.7\text{ k}\Omega \rightarrow 56\text{ k}\Omega$. $\ln$ = natural log (the “ln” key). Reference points: 63% at 1τ , 86% at 2τ , 95% at 3τ , 99% at 5τ .

DIRECT FORMULA

$$t = RC \times \ln(V_s / (V_s - V))$$

To half the supply: $t = 0.69 \times RC$

Problem 10 — Choose R or C for a filter, time constant or 555 timer

GIVEN target cutoff f_c , time constant τ , 555 frequency f or pulse t , and one chosen component.

FIND the other component.

1	$C = 1 / (2\pi \times R \times f_c)$	RC filter: from $f_c = 1/(2\pi RC)$. Swap R and C to find R .
2	$C = \tau / R$	From $\tau = R \times C$.
3	$R2 = (1.44 / (f \times C) - R1) / 2$	555 astable: from $f = 1.44 / ((R1 + 2R2) \times C)$.
4	$R = t / (1.1 \times C)$	555 monostable: from $t = 1.1 \times R \times C$.

WORKED EXAMPLE

1 kHz low-pass, $R = 10\text{ k}\Omega \rightarrow C = 1 / (6.283 \times 10\,000 \times 1000) = 15.9\text{ nF} \rightarrow 15\text{ nF}$ ($f_c = 1.06\text{ kHz}$).
555 at 1 kHz, $C = 10\text{ nF}$, $R1 = 1\text{ k}\Omega \rightarrow R2 = (144\text{ k}\Omega - 1\text{ k}\Omega) / 2 = 71.5\text{ k}\Omega \rightarrow 68\text{ k}\Omega$ ($f \approx 1.05\text{ kHz}$).
555 one-shot 5 s, $C = 100\text{ }\mu F \rightarrow R = 5 / (1.1 \times 0.0001) = 45.5\text{ k}\Omega \rightarrow 47\text{ k}\Omega$ (5.2 s).

DIRECT FORMULA

Pick C from the standard values first (fewer choices), then calculate R .
Electrolytic capacitors have $\pm 20\%$ tolerance — for accurate timing use film or C0G capacitors.

Problem 11 — Choose L or C for a resonant frequency (LC tank, Colpitts)

GIVEN target frequency f and one of L or C .

FIND the other component.

1	$C = 1 / ((2\pi f)^2 \times L)$	From $f = 1/(2\pi\sqrt{LC})$.
2	$L = 1 / ((2\pi f)^2 \times C)$	Same equation solved for L .
3	$C_{\text{tank}} = C1 \times C2 / (C1 + C2)$	Colpitts: the two feedback capacitors are in series in the tank.
4	$C1 = C2 = 2 \times C_{\text{tank}}$	Equal-capacitor choice (ratio $C1/C2$ sets the feedback).

WORKED EXAMPLE

7 MHz with $L = 10\text{ }\mu H \rightarrow (2\pi \times 7 \times 10^6)^2 = 1.93 \times 10^{15} \rightarrow C = 1 / (1.93 \times 10^{15} \times 10 \times 10^{-6}) = \mathbf{51.7\text{ pF}}$.
Colpitts: $C1 = C2 \approx 103\text{ pF} \rightarrow$ use 100 pF plus a small trimmer. Stray capacitance ($\approx 5\text{--}10\text{ pF}$) pulls f lower.

DIRECT FORMULA

$$C = 1 / (4\pi^2 f^2 L) \quad L = 1 / (4\pi^2 f^2 C)$$

Shortcut: C (pF) = 25 330 / ($f^2 \times L$), with f in MHz and L in μH .

Problem 12 — Find unknown C or L from an AC measurement

GIVEN AC voltage V and current I (RMS, from a multimeter), frequency f.

FIND capacitance C or inductance L.

1	$X = V / I$	Ohm's law with reactance instead of resistance.
2	$C = 1 / (2\pi \times f \times X_C)$	From $X_C = 1/(2\pi fC)$.
3	$L = X_L / (2\pi \times f)$	From $X_L = 2\pi fL$.
4	$X = \sqrt{Z^2 - R^2}$	If the part has resistance R (e.g. coil wire): $Z = V / I$, then remove R.

WORKED EXAMPLE

Capacitor on 12 V AC, 50 Hz, current 37.7 mA $\rightarrow X_C = 318 \Omega \rightarrow C = 1 / (314.16 \times 318) = \mathbf{10 \mu F}$.

Coil at 1 kHz, 1 V, 15.9 mA $\rightarrow X_L = 62.9 \Omega \rightarrow L = 62.9 / 6283 = \mathbf{10 mH}$. Works for sine waves only; use V and I as RMS values.

17.4 LEDs, Zener diodes and power supplies

Problem 13 — Several LEDs from one supply

GIVEN supply V_s , LED forward voltage V_f , LED current I , number of LEDs n in series.

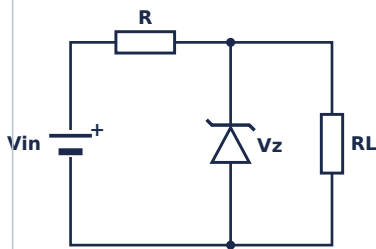
FIND series resistor R , its power, and the maximum number of LEDs.

1	$R = (V_s - n \times V_f) / I$	KVL: the LEDs take $n \times V_f$, the resistor takes the rest.
2	$P_R = (V_s - n \times V_f) \times I$	Heat in the resistor.
3	$n_{\max} = (V_s - 1.5 \text{ V}) / V_f$	Round down. Keep $\geq \sim 1.5 \text{ V}$ across R so current stays stable.
4	Parallel strings: one resistor per string	V_f differs between LEDs; a shared resistor makes one string hog the current.

WORKED EXAMPLE

12 V, 3 white LEDs (3.1 V), 20 mA $\rightarrow R = (12 - 9.3) / 0.02 = 135 \Omega \rightarrow \mathbf{150 \Omega}$ ($I = 18 \text{ mA}$). $P = 2.7 \text{ V} \times 0.018 \text{ A} = 0.05 \text{ W}$.
 $n_{\max} = (12 - 1.5) / 3.1 = 3$.

Problem 14 — Zener voltage regulator



GIVEN input range $V_{in,\min}$ – $V_{in,\max}$, Zener voltage V_z , maximum load current I_L , minimum Zener current $I_{z,\min}$ ($\approx 5 \text{ mA}$).

FIND series resistor R and the power ratings of R and the Zener.

1	$R = (V_{in,\min} - V_z) / (I_L + I_{z,\min})$	Worst case: lowest input, highest load. Round R down.
2	$I_{\max} = (V_{in,\max} - V_z) / R$	Highest current through R .
3	$P_z = V_z \times I_{\max}$	Worst case for the Zener: no load, all current goes through it.
4	$P_R = (V_{in,\max} - V_z)^2 / R$	Worst case for the resistor.

WORKED EXAMPLE

9–12 V in, 5.1 V Zener, 20 mA load $\rightarrow R = (9 - 5.1) / 0.025 = 156 \Omega \rightarrow \mathbf{150 \Omega}$.

$I_{\max} = 6.9 / 150 = 46 \text{ mA} \rightarrow P_z = 0.23 \text{ W} \rightarrow 0.5 \text{ W Zener}$. $P_R = 0.32 \text{ W} \rightarrow 1 \text{ W resistor}$ (2× margin). Wasteful above $\sim 50 \text{ mA}$ — use a linear or switching regulator instead.

Problem 15 — Transformer, rectifier and smoothing capacitor

GIVEN mains voltage V_p , transformer secondary V_s (RMS), DC load current I , allowed ripple ΔV .

FIND turns ratio, DC voltage after the rectifier, and smoothing capacitance.

1	$N_p / N_s = V_p / V_s$	Voltage scales with the turns ratio; current scales inversely: $I_p = I_s \times N_s / N_p$.
2	$V_{\text{peak}} = V_s \times \sqrt{2} - 1.4 \text{ V}$	Bridge rectifier: two diodes conduct at a time (use -0.7 V for one diode).
3	$C = I / (f_r \times \Delta V)$	Ripple formula $\Delta V = I / (f_r \times C)$. $f_r = 100 \text{ Hz}$ full-wave, 50 Hz half-wave (50 Hz mains).
4	$V_{\text{peak}} - \Delta V \geq V_{\text{out}} + \text{dropout}$	The ripple trough must stay above what a following regulator needs.

WORKED EXAMPLE

230 V $\rightarrow$ 12 V transformer: ratio 19.2 : 1. $V_{\text{peak}} = 12 \times 1.414 - 1.4 = 15.6 \text{ V}$. 0.5 A load, 1.5 V ripple $\rightarrow C = 0.5 / (100 \times 1.5) = 3333 \mu\text{F} \rightarrow \mathbf{4700 \mu\text{F}, \geq 25 \text{ V}}$. Trough 14.1 V $\geq 5 \text{ V} + 2 \text{ V}$ for a 7805 ✓. Size the transformer at 1.5–2× the DC power. Mains is lethal — work on the low-voltage side only.

Problem 16 — Heat in a linear regulator or transistor (heatsink sizing)

GIVEN power dissipated P , ambient temperature T_a , datasheet $T_{j,max}$, θ_{JA} , θ_{JC} .

FIND junction temperature, and whether a heatsink is needed and how good it must be.

1	$P = (V_{in} - V_{out}) \times I$	Linear regulator: the voltage difference is burned as heat. For transistors use P_D from sections 7–8.
2	$T_j = T_a + P \times \theta_{JA}$	Without heatsink. θ is temperature rise per watt ($^{\circ}\text{C}/\text{W}$).
3	$\theta_{total} = (T_{j,max} - T_a) / P$	Maximum allowed thermal resistance, junction to air.
4	$\theta_{SA} = \theta_{total} - \theta_{JC} - \theta_{CS}$	Required heatsink rating. $\theta_{CS} \approx 0.5 \text{ }^{\circ}\text{C}/\text{W}$ with thermal paste. Pick a heatsink with a lower θ_{SA} .

WORKED EXAMPLE

7805 from 12 V at 0.5 A $\rightarrow P = 7 \times 0.5 = 3.5 \text{ W}$. TO-220 without heatsink ($\theta_{JA} \approx 50 \text{ }^{\circ}\text{C}/\text{W}$): rise $175 \text{ }^{\circ}\text{C} \rightarrow$ too hot.

$T_{j,max}$ 125 $^{\circ}\text{C}$, T_a 40 $^{\circ}\text{C} \rightarrow \theta_{total} = 85 / 3.5 = 24.3 \rightarrow \theta_{SA} = 24.3 - 5 - 0.5 = \leq \mathbf{18.8 \text{ }^{\circ}\text{C}/\text{W} \text{ heatsink}}$. Or use a switching (buck) regulator and avoid most of the heat.

17.5 Transistor circuits

Problem 17 — BJT switch for a load (relay, motor, LED strip) from a GPIO pin

GIVEN load supply V_{cc} , load resistance or current, GPIO voltage, transistor $h_{FE(min)}$.

FIND collector current, base current and base resistor R_B .

1	$I_C = (V_{cc} - V_{CE(sat)}) / R_{load}$	Current the load draws when the transistor is fully on.
2	$I_B = I_C / 10$	Forced gain of 10 guarantees saturation (h_{FE} varies between parts).
3	$R_B = (V_{gpio} - 0.7) / I_B$	Round down to a standard value (slightly more base current is fine).
4	Check: $I_B \leq \text{pin limit}$, $I_C \leq I_{C(max)}$	Inductive loads need a flyback diode across the load, cathode to $+V_{cc}$.

WORKED EXAMPLE

12 V relay, coil 400 Ω , 3.3 V GPIO $\rightarrow I_C = (12 - 0.2) / 400 = 29.5 \text{ mA} \rightarrow I_B = 2.95 \text{ mA} \rightarrow R_B = 2.6 / 0.00295 = 881 \Omega \rightarrow \mathbf{820 \Omega}$ ($I_B = 3.2 \text{ mA}$, within pin limits). Add a 1N4148 across the coil.

Problem 18 — BJT amplifier bias point (voltage-divider bias)

GIVEN V_{cc} , base divider R1 (top) and R2 (bottom), emitter resistor R_E , collector resistor R_C .

FIND the DC voltages and currents — to check the transistor sits in its active region.

1	$V_B = V_{cc} \times R2 / (R1 + R2)$	Valid when divider current $\geq 10 \times I_B$ (base current barely loads it).
2	$V_E = V_B - 0.7 \text{ V}$	Base-emitter junction drop.
3	$I_C \approx I_E = V_E / R_E$	R_E sets the current — independent of h_{FE} .
4	$V_C = V_{cc} - I_C \times R_C$; $V_{CE} = V_C - V_E$	Active region if $V_{CE} > \sim 1 \text{ V}$. Best swing when V_C is near the middle.

WORKED EXAMPLE

$V_{cc} = 12 \text{ V}$, $R1 = 47 \text{ k}\Omega$, $R2 = 10 \text{ k}\Omega$, $R_E = 1 \text{ k}\Omega$, $R_C = 4.7 \text{ k}\Omega \rightarrow V_B = 2.11 \text{ V} \rightarrow V_E = 1.41 \text{ V} \rightarrow I_C = 1.41 \text{ mA} \rightarrow V_C = 12 - 6.6 = 5.4 \text{ V} \rightarrow V_{CE} = \mathbf{4.0 \text{ V}}$ ✓ active. Voltage gain $\approx -R_C / R_E = -4.7$ (with R_E not bypassed).

Problem 19 — Check a MOSFET switch

GIVEN load current I_D , gate drive voltage, datasheet $R_{DS(on)}$ at that V_{GS} and θ_{JA} .

FIND power loss and temperature rise.

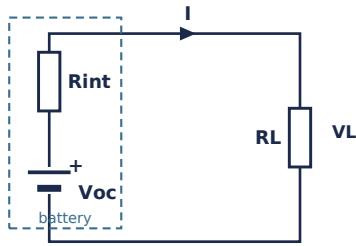
1	$R_{DS(on)}$ at your V_{GS}	Read the value specified for your drive voltage (3.3 V drive needs a part rated at 2.5 V).
2	$P = I_D^2 \times R_{DS(on)}$	Conduction loss; convert m Ω to Ω .
3	$\Delta T = P \times \theta_{JA}$	Temperature rise above ambient (see Problem 16).
4	Gate: 10-100 Ω series, 10-100 k Ω gate-to-source	The pull-down keeps the MOSFET off while the MCU boots.

WORKED EXAMPLE

3 A load, $R_{DS(on)} = 40 \text{ m}\Omega$ at 3.3 V $\rightarrow P = 9 \times 0.04 = 0.36 \text{ W}$. SOT-23 package, $\theta_{JA} \approx 125 \text{ }^\circ\text{C/W} \rightarrow \Delta T = 45 \text{ }^\circ\text{C} \rightarrow$ acceptable at room temperature; at 5 A ($P = 1 \text{ W}$, $\Delta T = 125 \text{ }^\circ\text{C}$) choose a larger package or lower $R_{DS(on)}$.

17.6 Batteries and sources

Problem 20 — Measure a battery's internal resistance



GIVEN open-circuit voltage V_{oc} (no load), voltage V_L with a known load R_L connected.

FIND internal resistance R_{int} , then voltage sag at any current.

1 $I = V_L / R_L$

Current drawn by the known load.

2 $R_{int} = (V_{oc} - V_L) / I$

The missing voltage is dropped inside the battery.

3 $V = V_{oc} - I_{new} \times R_{int}$

Predict terminal voltage at another current.

WORKED EXAMPLE

$V_{oc} = 4.10 \text{ V}$, with 10Ω : $V_L = 3.95 \text{ V} \rightarrow I = 0.395 \text{ A} \rightarrow R_{int} = 0.15 / 0.395 = \mathbf{0.38 \Omega}$.

At 2 A : $V = 4.10 - 0.76 = 3.34 \text{ V}$ — a 3.3 V regulator would drop out.

Measure quickly; voltage recovers after the load is removed.

DIRECT FORMULA

$$R_{int} = R_L \times (V_{oc} - V_L) / V_L$$

Same method works for any supply, power bank or long cable.

Problem 21 — Battery packs: cells in series and parallel

Connection	Voltage	Capacity (Ah)	Energy (Wh)	Internal resistance
Series (S)	Adds: $n \times V_{cell}$	Same as one cell	Adds	Adds: $n \times R_{cell}$
Parallel (P)	Same as one cell	Adds: $n \times Ah_{cell}$	Adds	Divides: R_{cell} / n
xS yP pack	$x \times V_{cell}$	$y \times Ah_{cell}$	$x \times y \times Wh_{cell}$	$R_{cell} \times x / y$

Example: 3S2P of 3.7 V , 2.5 Ah cells $\rightarrow 11.1 \text{ V}$, 5 Ah , 55.5 Wh . At 1 A load $\rightarrow$ runtime $\approx 5 \text{ h}$ (expect $\approx 4 \text{ h}$). Only parallel cells of the same type and charge level; series Li-ion packs need a BMS to balance the cells.

Quick reference only — check results against real measurements and datasheets. Standard resistor values (E12): 10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82 $\times$ powers of 10.